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Invited Review

Resource-constrained multi-project scheduling problem: A survey

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ABSTRACT

Project Management is becoming increasingly crucial in competitive environments such as manufacturing and the service industries. The Resource-Constrained Multi-Project Scheduling Problem (RCMPSP) consists of assigning start times to jobs corresponding to two or more projects that must be executed simultaneously while respecting the precedence between jobs and limited resources. The existing rise in the study of the RCMPSP resulted in numerous works on the topic while proposing different problem features. This research analyzes different variants of the problem based on aspects related to jobs, projects, relationships, resources, and time management. Moreover, based on the problem variants considered in the collected works, a taxonomy allowing (i) the identification and positioning of each RCMPSP variant and (ii) the analysis of the current state-of-the-art of the problem is proposed. In addition, the solution methods proposed to solve RCMPSPs are classified and analyzed, together with the benchmarks used to validate such approaches. Finally, this review paper discusses the RCMPSP connection to practice and provides future research opportunities in light of recent advances.

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1. Introduction

Project scheduling problems constitute, within the field of Operations Research, one of the most important problems for both research and practice. The planning of the project schedule is considered by Pinto & Prescott (1990) as one of the essential success factors in project planning. The first ideas related to this problem emerged at the beginning of the last century when the Gantt charts started to be used in Gantt (1913). Later in the 1930s, the study of problems related to Job Shop Scheduling was started (Jain & Meeran, 1999). In the 1950s, methods for the control and planning of projects such as program evaluation and review techniques (Malcolm, Roseboom, Clark, & Fazar, 1959) and the critical path method (Kelley & Walker, 1959) emerged. Studies and techniques developed to date were based on the existence of almost infinite resources. However, the development of industrial processes and the increase in complex production and consumption lines led, at the beginning of the 1960s, to the study of planning and control of projects taking into consideration the existence of limited resources. That resulted in the proposition of the resource-constrained project-scheduling problem (RCPSP) by Dike (1964) which was demonstrated to be NP-hard by Lenstra & Rinnooy Kan (1978). The RCPSP consists of scheduling processing start times of various jobs that are part of a project optimizing a given objective function while respecting existing resource availability and precedence restrictions. That is, a job can only start executing when its predecessor jobs have been completed and there is enough resource availability. This problem has been widely studied in the literature in research such as Kong & Dou (2021), Wang, Hu, Demeulemeester, & Zhao (2021), Zaman, Elsayed, Sarker, Essam, & Coello Coello (2021), Laurent, Deroussi, Grangeon, & Norre (2017), Chen (2011), Coelho & Vanhoucke (2011), Voß & Witt (2007), and Brucker, Drexl, Möhring, Neumann, & Pesch (1999). Recently, Hartmann & Briskorn (2022) developed a follow-up survey to Hartmann & Briskorn (2010); in both works, the authors presented an overview of variants of the RCPSP.

As an extension of the RCPSP, Pritsker, Watters, & Wolfe (1969) presented the resource-constrained multi-project scheduling problem (RCMPSP), where several projects must be executed simultaneously, competing for the use of resources. That extension also involved adding the arrival times of each project to the production line. In this sense, the RCMPSP has been one of the most studied and implemented extensions at the business level in dif-

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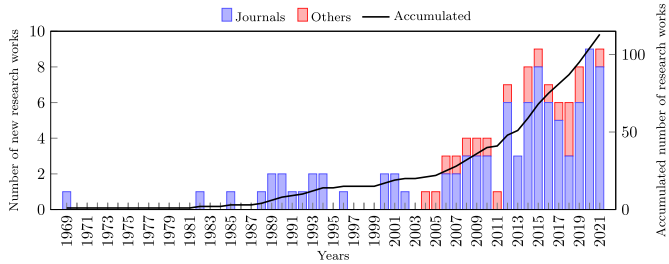

Fig. 1. Distribution per year of research works that address the RCMPSP collected in the current document.

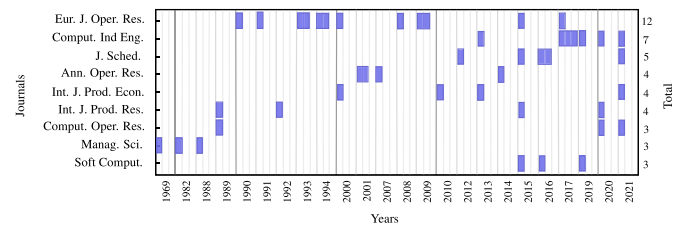

Fig. 2. Distribution of the number of publications collected in the current document by journal.

ferent industry sectors. According to data presented by Lova & Tormos (2001), at the beginning of this century, 84% of the companies worldwide were working on multiple projects simultaneously, where efficient and effective management of resources was highlighted as a key factor for the long-term success of companies. In addition, the investigation of Karaa & Nasr (1986) indicated that the efficient management of resources provides a competitive advantage and added value to those companies considering it. Due to this, the study of the RCMPSP presented a considerable increase in the number of research works, especially in the past and current decade (see Fig. 1). Also, this resulted, besides algorithms and solution approaches, in multiple RCMPSP variants.

The works Hartmann & Briskorn (2022) and Hartmann & Briskorn (2010) addressed a wide scope of RCPSP variants, including the RCMPSP. Given the vast number of related research from the different RCPSP variants, the authors only addressed a part of the research on the RCMPSP. Thus, there were several features that, to the best of our knowledge, are not described in the existing review works due to their understandable scope, for example, mobility of resources and uncertainty in several aspects. On the other hand, Issa & Tu (2020) presented another survey on the RCPSP and the RCMPSP, which provides a classification in terms of flexibility concerning the resource use, duration, and preemption of the jobs. However, that survey only considered a very small number of RCMPSP-related works. Due to this, several features and objective functions remain unidentified and not analyzed, e.g., job and project quality, cost and time of resource transfers, local resources, total project profit maximization, and time balance maximization, among others.

The contribution of this paper is three-fold: (i) a review of the literature analyzing and discussing features and objective functions proposed so far for the RCMPSP as well as solution methods and existing benchmarks; (ii) a taxonomy that classifies different aspects related to the problem, providing a better understanding of the current state of the problem; and (iii) an analysis of the main trends in the last years and the problem application in real environments, in order to guide towards future research lines.

The remainder of this paper is organized as follows. In Section 2, the methodology used to carry out this research is described. In Section 3, the description of the problem in its simplest form and an illustrative example are presented. That section, thus, sets the bases for analyzing problem variants in Section 4. The solution methods and benchmarks used to address the RCMPSP and variants are exposed in Sections 5 and 6, respectively. The RCMPSP connection to practice is discussed in Section 7 and, finally, in Section 8 the conclusions of the research and several lines of future research are presented. In the Appendix A, a glossary of abbreviations used in this article is provided.

2. Survey methodology

This section provides a description of the conducted methodology for collecting and processing related works. After related works were collected, 34 problem features and 20 objective functions were identified based on which the proposed taxonomies were formed. Moreover, the design of this survey is oriented toward the analysis of the RCMPSP and related variants through the detailed study of each included paper in terms of problem features, solution approaches, proposed benchmarks, and analysis by means of various statistics, e.g., more studied aspects, relevant topics in recent years, and connection to practice.

The scientific literature collected in this review corresponds to manuscripts indexed in *Scopus*, *WoS*, considering also publishers *IEEE Xplore*, *ScienceDirect by Elsevier*, *Wiley* and *SpringerLink*. The search process was based on (i) manuscripts written in the English language, (ii) including journal papers, conferences, and review papers, and (iii) searching for keywords in title, abstracts, and keywords. The used keywords were *resource-constrained*[1], *multi-project*[1], *project management, RCMPSP, RC-MPSP, and scheduling problem*. The full texts of the obtained manuscripts were analyzed to filter the papers that addressed the RCMPSP or related variants. The collected literature is scattered from 1969 to 2021, and the majority corresponds to journal articles in the areas of operations research, artificial intelligence, management science, and scheduling. The journals with most RCMPSP related works are the *European Journal of Operational Research*, *Computers & Industrial Engineering*, *Journal of Scheduling*, etc. Figure 2 shows the amount of research related to the RCMPSP carried out per year in the main journals. The width of the horizontal bars indicates the number of related papers published by year and journal (between 0 and 3). The total number of papers published in each journal from 1969 to 2021 is indicated on the right side of the figure. Only journals where at least three works were published are shown.

3. Problem statement

In this work, we refer to the *base problem* as the most simplified RCMPSP variant, composed of a set of projects, each composed of a subset of jobs. In order to complete jobs, there is a constant number of shared renewable resources available in each time period. Moreover, each project defines a release date or arrival time, that is, the time from which the jobs can start and the desired date that represents the desired time period in which the project should be completed. Each job corresponding to the project is defined by a processing time and a given quantity required of each resource type. Some jobs belonging to the same project might establish end-start precedence relationships, i.e., a job can only start after a set of predecessor jobs are completed. A feasible schedule for the problem consists of assigning start times to each job in each project, respecting the precedence relationships between jobs, and the availability of resources at each time period. Table 1 shows the existing works in the reviewed literature that address the base problem. The base problem has been studied by pursuing several objective functions (see Section 4.2), with the minimization of the time at which all projects are completed or the makespan (M) being the most used one.

[1] The search was performed using those words hyphenated and non-hyphenated.

Table 1
Research works collected in the current document that address the base problem.

Van Eynde & Vanhoucke (2020), Chakrabortty et al. (2017), Pérez et al. (2016), Gonçalves et al. (2015), Sonmez & Uysal (2015), Vázquez et al. (2015), Zhang & Sun (2011), Browning & Yassine (2010b), Gonçalves et al. (2008), Linyi & Yan (2007), Zhuang & Yassine (2004), Lova & Tormos (2001), Lova et al. (2000), Tsai & Chiu (1996), Lawrence & Morton (1993), Dumond (1992), Deckro, Winkofsky, Hebert, & Gagnon (1991), Tsubakitani & Deckro (1990), Dumond & Mabert (1988), Kurtulus & Davis (1982)

3.1. Mathematical formulation

In this section, the mathematical model of the base problem aiming at minimizing the makespan is presented. The parameters of the problem are the following:

I	Set of projects,
i	Project number, $i \in \{1, 2, \ldots, \lvert I\rvert\}$,
N_i	Set of jobs in project i,
j	Job number, $j \in \{1, 2, \ldots, \lvert N_i\rvert\}$,
AD_i	Absolute due date (i.e., hard deadline) of project i,
DD_i	Desired due date of project i,
AT_i	Arrival time of project i,
t	Time period, $t \in \{1, 2, \ldots, max(AD_1, AD_2, \ldots, AD_{\lvert I\rvert})\}$,
K	Set of different resource types,
k	Resource number, $k \in \{1, 2, \ldots, \lvert K\rvert\}$,
P_{ij}	Set of jobs being predecessors of job j of project i,
d_{ij}	Number of periods required to perform job j of project i,
c^k_{ij}	Consumption of resource type k required to perform job j of project i,
a_k	Availability of resource k at every time period.

From AT_i, AD_i, P_{ij} and d_{ij} and assessing that there are no resource constraints, the following parameters, exemplified by Pritsker et al. (1969), can be calculated:

L_{ij}	Earliest possible period in which job j of project i could be completed,
U_{ij}	Latest possible period in which job j of project i could be completed.

The following decision variable is defined:

$$x^t_{ij} = \begin{cases} 1, & \text{if job } j \text{ of project } i \text{ is completed in period } t, \\ 0, & \text{otherwise.} \end{cases}$$

The constraints established for the base problem are expressed as follows:

$$\sum_{t=L_{ij}}^{U_{ij}} x^t_{ij} = 1, \quad \forall\, i \in I, j \in N_i \tag{1}$$

$$\sum_{t=L_{ij}}^{U_{ij}} t x^t_{ij} - d_{ij} \geq \sum_{t=L_{im}}^{U_{im}} t x^t_{im}, \quad \forall\, i \in I, j \in N_i, m \in P_{ij} \tag{2}$$

$$\sum_{i=1}^{\lvert I\rvert} \sum_{j=1}^{\lvert N_i\rvert} \sum_{q=t}^{t+D_{ij}-1} c^k_{ij} x^q_{ij} \leq a_k, \quad \forall\, k \in K, t \in max(AD_1, AD_2, \ldots, AD_{\lvert I\rvert}) \tag{3}$$

$$x^t_{ij} \in \{0, 1\}, \quad \forall\, i \in I, j \in N_i, t \in max(AD_1, AD_2, \ldots, AD_{\lvert I\rvert}) \tag{4}$$

Constraints (1) ensure that each job has only one completion time. Constraints (2) establish the relationship between jobs, where a job cannot start its execution before the jobs of its set of predecessor jobs are completed. Constraints (3) ensure that only available resources are consumed at each time period. Constraints (4) establish the definition of the decision variables.

The minimization of the makespan can be expressed as follows:

$$\text{Minimize } \; max(t x^t_{ij}), \; i \in I, j \in N_i, t \in L_{ij}, \ldots, U_{ij} \tag{5}$$

Table 2
Data for the illustrative example of the base problem for two projects.

Jobs	d_{ij}	P_{ij}	c^1_{ij}	c^2_{ij}	c^3_{ij}
Job_{11}	3	-	5	-	-
Job_{12}	5	Job_{11}	5	8	-
Job_{13}	4	Job_{12}	3	-	-
Job_{14}	3	Job_{11}	-	3	7
Job_{21}	5	-	-	1	-
Job_{22}	4	Job_{21}	-	3	-
Job_{23}	4	Job_{21}	-	-	3

3.2. RCMPSP solution example

We present an illustrative example of the base problem. There are two projects: $Project_1$, composed of the jobs Job_{11}, Job_{12}, Job_{13}, and Job_{14}; and $Project_2$, composed of the jobs Job_{21}, Job_{22}, and Job_{23}. The time corresponding to AT_i is 0 and 2 for $Project_1$ and $Project_2$, respectively. There are three types of renewable and global resources with availability $A_1 = 10$, $A_2 = 9$, and $A_3 = 11$. Table 2 shows the duration, predecessor jobs, and resource consumption of each job in each project.

The graphical representation of several projects sharing global resources using the activity-on-node network can be carried out using the single- or multi-project approach. Both approaches use starting and ending dummy nodes known as *super-source* and *super-sink*. In the single-project approach, there is a starting dummy node and an ending dummy node representing the start and completion of all projects, respectively (see Fig. 3(a)). This representation cannot be used when each project pursues its objective function; for example, each project aims to minimize its makespan regardless of the progress of the other projects (e.g., Li, Xu, & Li, 2021). On the other hand, in the multi-project approach, each project includes a starting dummy node and an ending dummy node (see Fig. 3(b)). This representation can be used when both individual goals and a common goal are pursued.

In the illustrative example, Job_{12} and Job_{14} can be executed from time period 3, if only the precedence relationships are considered (see Table 2, Column 3). Since in the problem, there are restricted resources and the required consumption of the second resource by both jobs ($C_{12} + C_{14} = 8 + 3 = 11$) is greater than the availability of the resource in each time period ($A_2 = 9$) both jobs cannot be executed simultaneously. This situation is reflected between several jobs and even different projects; for instance, between Job_{12} and Job_{22}, where there is no established relationship of precedence. Figure 4 presents two feasible solutions to the problem.

In Fig. 4(a), the completion time of Job_{14} (last job executed belonging to $Project_1$) is 15 and in Fig. 4(b), the completion time of Job_{13} and Job_{22} (last jobs executed belonging to $Project_1$ and $Project_2$) is 12. The solution in Fig. 4(b) presents the lowest possible makespan value; therefore, it represents the optimal solution for the illustrative example.

4. A review of RCMPSP variants

This section presents a summary of the RCMPSP variants studied in the literature. Given the nature of the problem and collected papers, variants are analyzed in terms of features and objective functions, presenting classifications for both aspects.

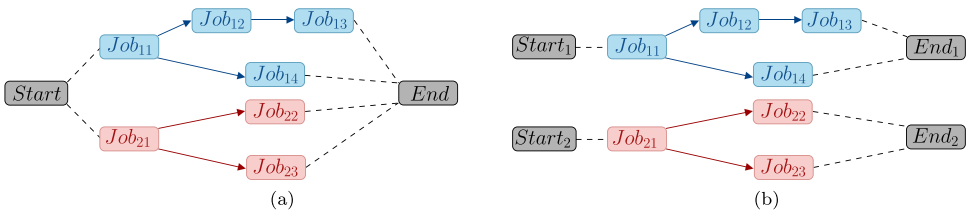

Fig. 3. (a) Single-project approach, and (b) Multi-project approach.

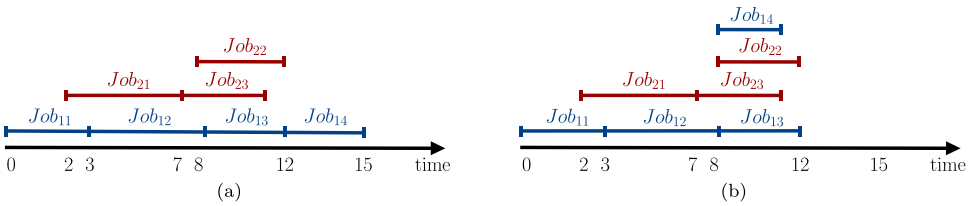

Fig. 4. (a) Feasible solution, and (b) Optimal solution.

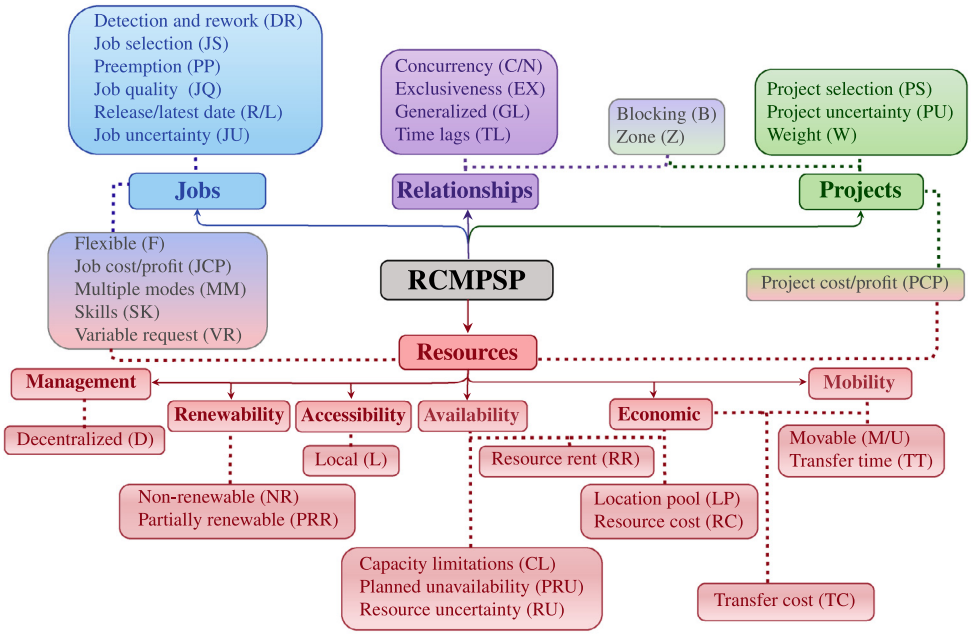

Fig. 5. Classification of the RCMPSP variants.

4.1. Problem features

In order to classify the features found in the studied manuscripts, four main components are defined: jobs, relationships, projects, and resources. Figure 5 shows a taxonomy based on all RCMPSP variants proposed. The colors blue, purple, green, and red represent features related to jobs, relationships, projects, and resources, respectively.

Several features studied in the single-project environment are also present in multi-project environments, e.g., multiple execution modes, non-renewable resources, and generalized relationships. On the other hand, some other features are only specific to multi-project cases, e.g., decentralized management, local resources, and project selection, as well as different objectives for different projects. In this review, all features found in the literature appearing in RCMPSP works are described and those only appearing in multi-project environments are further elaborated.

4.1.1. Job- and resource-based features

One of the essential conditions for starting a job execution is the availability of the required resources. The existence of multiple execution modes (MM) is one of the well-known related features, where a different duration and resource consumption denote each possible job execution mode. Generally, a higher resource consumption implies shorter duration. There may be a set of jobs that need to be executed in the same mode, but commonly the job execution modes are independent. As an illustrative example, consider that cleaning an area takes 30 minutes if only one machine is used and 15 minutes if two machines are used. A particular case of multiple execution modes involves the use of the workforce as a resource in project planning, where each person/worker presents skills or efficiency levels to perform one or more jobs (SK). When all associated parameters are known, e.g., duration and resource consumption of each job when performed by each worker, the situation can be modeled as multiple execution modes. On the other hand, in some cases, the mentioned parameters must be calculated

Table 3
Research works that address job- and resource-based features.

Features	Research works
Job cost/profit (JCP)	He et al. (2021), Xu & Feng (2014), Zapata et al. (2008), Kolisch & Meyer (2006), Chiu & Tsai (2002)
Multiples modes (MM)	Davari Ardakani & Dehghani (2022), Ahmeti & Musliu (2021), Araujo et al. (2020), Kannimuthu et al. (2020), Cheng et al. (2019), Zhang & Chen (2018), Geiger (2017), Joo & Chua (2017), Asta et al. (2016), Toffolo, Santos, Carvalho, & Soares (2016), Wauters et al. (2016), Beşikci, Bilge, & Ulusoy (2015), Chen & Ju (2015), Can & Ulusoy (2014), Chen, Zhu, & Zhang (2014), Xu & Feng (2014), Ju & Chen (2012), Zapata et al. (2008), Kumanan, Jose, & Raja (2006), Kolisch & Meyer (2006), Vercellis (1994), Speranza & Vercellis (1993), Pritsker et al. (1969)
Skills (SK)	Cui, Liu, Lu, & Jia (2021), Namazian & Yakhchali (2016), Chen et al. (2014), Kolisch & Heimerl (2012), Gutjahr, Katzensteiner, Reiter, Stummer, & Denk (2008)

depending on the efficiency or skill level the person/worker has to perform each job and other aspects such as resource availability or required quality. Considering multiple skills or efficiency levels could imply changes in the job duration, resource consumption, or job completion quality.

A similar feature to multiple execution modes is the flexible resource allocation to a job (F) studied by Issa, Patterson, & Tu (2021) and Rokou, Dermitzakis, & Kirytopoulos (2014). This involves that executing a certain job requires a total resource amount, i.e., workload, so the resource allocation in each time period is not necessarily following a uniform distribution. In this sense, for each job, a set of modes can be automatically defined, composed of the possible combinations of duration and resource required. Each defined mode for each job must comply with the required resource amount. Closely related to the previously described features, Davari Ardakani & Dehghani (2022) and Kurtulus (1985) analyzed the variable resource requirement (VR). In this case, the job resource requirements vary at each time period but are initially defined; for example, to start executing a job in a machine, two workers are necessary, but once the execution process is stabilized, only one worker is required to complete the job execution.

In a production process, cash or liquidity can be seen as a resource, so the start of job execution represents a cash outflow, e.g., service payments, while the completion of job execution represents a cash inflow, e.g., the sale of products. This job cost/profit (JCP) was studied in several research works and, when considered in a problem, maximizing the net present value is the objective function most frequently considered.

Table 3 shows the existing research in the reviewed literature that addresses job- and resource-based features. Note that only features considered in at least five works are listed, the other cases are referenced in this section.

4.1.2. Job-based features

In the base problem, the jobs follow certain primary conditions, for example, deterministic duration and continuous execution. These aspects often move the problem away from real production processes where these features can be modified.

Continuous job execution involves a few outdated production procedures. In this sense, the concept of preemption (PP) arises when job execution interruption is allowed due to job preferences. Job execution splitting can lead to more stable resource use by realizing consumption in time periods where resources could remain idle. This feature can be accompanied by costs associated with stopping and resuming the job execution, which can lead, for example, to an analysis of cost versus completion time. The handling of uncertainty always brings a complexity increase in any problem resolution. Despite this, several works studied the existing uncertainty on job duration (JU) since unexpected events affecting job duration might occur in real processes.

Job release and/or latest date (R/L) studied by Joo & Chua (2017) and Hao, Shen, Xue, & Wang (2010) can have a relevant influence on the schedule. This feature implies that it is impossible to start the job execution before the established release date or finish it after the established latest date. In the literature this concept is also known as time constraints or *start-no-earlier-than* and *start-no-later-than*. When problem conditions do not impose the job release date, the value is calculated based on the predecessors and duration of each job as well as the arrival time and absolute due date of the corresponding project.

As part of the study of multiple execution modes, Kannimuthu, Raphael, Ekambaram, & Kuppuswamy (2020) and Xu & Feng (2014) studied job quality (JQ), considering different quality levels for each execution mode. This concept can lead, for example, to an analysis of quality versus completion time or quality versus revenue when additional payments are established depending on the job quality. The quality can be determined depending on the project environment; for example, in construction projects, the quality can be calculated based on the construction quality assessment system (CONQUAS) (BCA, 2017). Detection/inspection and reworks (DR) was also proposed as a job-related variant in the research works presented by Zhu, Lu, Lu, & Ren (2021), Zhu, Lu, & Hu (2018), and Bock & Patterson (1990), where, after a project is finished, an error detection process is carried out and, faulty jobs are corrected. The existence of uncertainty when this feature is studied is implicit; therefore, it is not described as an independent feature.

The existence of projects that have a set of jobs whereof only one of them is executed was studied by Hauder, Beham, Raggl, Parragh, & Affenzeller (2020) under the concept of alternative activities. In the RCPSP literature, it is related to concepts of logical dependencies, or job selection (JS), the latter being the one used in this research work. The jobs can differ in several characteristics, such as duration, resource consumption, quality, and precedence relationships. When studying the problem under different execution modes for a job, the execution modes of each job can be seen as job selection, where the difference between each pair of jobs is based on the duration and resource requirements. Depending on other characteristics that can be added to the problem, this feature allows, for example, the analysis of the use of resources versus completion time.

Table 4 shows the existing research in the reviewed literature that addresses job-based features. Note that only features considered in at least five works are listed, the other cases are referenced in this section.

4.1.3. Resource-based features

Resource allocation is a significant aspect of the RCMPSP that adds a strong degree of complexity. Efficient use and availability of resources are a growing concern in society. As a consequence, there are several resource-based features, so eight sub-classifications are defined (i.e., Management, Renewability, Accessibility, Availability, Mobility, Economic, Availability and Economic, Economic, and Mobility), and 13 features are shown in Fig. 5.

- *Management*: In the RCMPSP, resource management can be centralized or decentralized. In the first case, only one general decision-maker manages the resources knowing the current status of each project and pursuing a global objective. In the other

Table 4
Research works that address job-based features.

Features	Research works
Preemption (PP)	Issa et al. (2021), Joo & Chua (2017), Hao et al. (2010), Bock & Patterson (1990), Pritsker et al. (1969)
Job uncertainty (JU)	Satic et al. (2022), Liu & Xu (2020), Nabipoor et al. (2020), Tian et al. (2020), Chen et al. (2019), Namazian et al. (2019), Song, Xi, Kang, & Zhang (2018), Wang et al. (2017), Hu et al. (2015), Xu & Feng (2014), Xu & Zhang (2012), Mao et al. (2009), Zapata et al. (2008), Kim & Schniederjans (1989)

case, when the resources are managed decentrally (D), the allocation of global resources is carried out by a general decision-maker who pursues a global objective but might not fully know the status of each project. Once this allocation is made, the decision-makers of each project are responsible for assigning execution start times to each job, complying with the resources assigned by the general decision-maker, and pursuing a local objective. Following Homberger (2007), each decision-maker plans exactly one project and is considered "honest" when providing information regarding the planning of its related project.

• *Renewability*: In management processes, there are different types of renewal of existing resources, e.g., renewable, non-renewable, or partially renewable resources. Renewable resources present a fixed existence that is renewed in each time period. Non-renewable resources (NR) arise from the need to optimize resources that present a unique existence for the entire time horizon so that once they are assigned to a job, they are not renewed, e.g., a unique budget. The study of non-renewable resources is closely related to multiple execution modes, given the different consumption of resources. Another related feature, studied by Amirian & Sahraeian (2017), refers to the existence of partially renewable resources (PRR), where the resources are renewed in a set period of time, e.g., a budget renewed several times within the time horizon.

• *Accessibility*: The different access to existing resources is a widely studied feature in the multi-project environment. The accessibility can be distinguished between global and local resources. The global resources can be used for the execution of all jobs of all projects. Otherwise, local resources (L) are defined as resources belonging to a project that can be consumed exclusively by the jobs belonging to the project, e.g., workers of a company branch office. In the papers studied, the resources are defined as only one category; that is, the access to a resource cannot be local and at the same time global. For instance, Fu & Zhou (2021) define local resources as fixed facilities and unskilled workers and global resources as specialized equipment and skilled professionals. Both accessibility options were also studied in relation to resource renewability. For example, Ahmeti & Musliu (2021), Asta, Karapetyan, Kheiri, Özcan, & Parkes (2016), and Wauters et al. (2016) studied local renewable and non-renewable resources, and global non-renewable resources. However, research presented by Geiger (2017) assumes that local resources are only non-renewable and global resources are considered only renewable. On the other hand, although local resources can be studied when working centrally, the beginning of their study associated with the RCMPSP is closely linked to decentralized management, where the decision-makers of each project are responsible for the management of its resources.

• *Availability*: The fixed resource availability at each time period has been considered since the beginning of the RCMPSPs. Studying different conditions where there may be uncertainty in the resource existence or even the non-existence of resources results in problem variants to cope with real situations in production lines. Uncertainty in resource existence (RU) moves the problem to a stochastic area, increasing resolution complexity, especially in large production processes. This aspect was studied by Hu et al. (2015), Wang, Chen, Mao, Chen, & Li (2015), Mao, Roos, & Salden (2009), and Zapata, Hodge, & Reklaitis (2008). Similarly, Tian, Dong, & Han (2018) studied resource unavailability planning (RUP). This feature provides flexibility to the resource availability, keeping the problem in a deterministic environment. In this case, unavailability planning refers to periods where there is no specific resource, e.g., break times for machines or workers. On the other hand, when the resources are transported to a project to be used later, the resource availability can be affected by capacity limitations (CL), for example, related to transportation and inventory. In the same way, the job completion can generate products or even waste that needs to be stored and transported later. This feature was studied by Gholizadeh-Tayyar, Dupont, Lamothe, & Falcon (2016) and Tayyar, Lamothe, & Dupont (2016) and is related to the concept of cumulative resources in the RCPSP.

• *Mobility*: This aspect defines the need to move some resources between jobs to carry out the executions. Mobility can arise between jobs of the same project and different projects. The transfer times (TT) involve all necessary movements for starting the job execution and can be used to model setup times, e.g., the time it takes to move a resource from one project to another for the execution process or the time it takes to move a job to the area where it can be executed. Transfer times can have a considerable influence when preemption is allowed. In the same way, Tian et al. (2018) considered that the resources could be movable or unmovable (M/U). In the movable case, the jobs are kept in an area, and the resources are transferred to jobs. In the un-movable case, the jobs must move to the resources to receive the requested service, e.g., big machines already fixed in a given space.

• *Economic*: The economic aspect is also reflected in some features found in the literature. The resource cost (RC) is a feature that several works address. Each resource is associated with a cost. That can lead, for example, to an analysis of cost versus completion time, especially when multiple execution modes are considered. In addition, the resource cost is also addressed when overtime is considered as a way of increasing production capacity. The costs in regular hours are usually lower than in overtime, which can lead, for example, to a trade-off analysis of cost for overtime versus cost for the delay. Another cost associated with resources can be inventory costs due to storage. Rostami & Bagherpour (2020) studied the need to designate a resource pool for doing periodic services to projects (LP), establishing that the location of the resource pool is not set. Instead, there are potential locations where the facility for the resource pool can be constructed. Given this, it is necessary to determine the appropriate location to construct the facility while incurring the lowest possible costs. The construction costs of the facility may vary depending on the selected location and the transportation costs of the resources from the facility to the different projects.

• *Mobility and economic*: As a consequence of the resource mobility, Fu & Zhou (2021), Suresh, Dutta, & Jain (2015), Adhau, Mittal, & Mittal (2013), and Krüger & Scholl (2010), consider that a transfer cost (TC) is also generated due to physically moving resources. Similarly, this cost can be used when coping with transfers between jobs of the same project, between different projects, or even between the resource pool and each project.

• *Availability and economic*: The restricted existence of renewable resources can cause timeouts for a job execution start, generating delays in a project completion and costs for delays. As a

Table 5
Research works that address resource-based features.

Features	Research works
Decentralized (D)	Li et al. (2021), Fu & Zhou (2021), Liu & Xu (2020), Rostami & Bagherpour (2020), Cheng et al. (2019), Jedrzejowicz & Ratajczak-Ropel (2019), Shi, Du, & Li (2019), Li & Xu (2018), Zhang & Chen (2018), Zheng, Guo, Zhu, & Zhang (2014), Adhau et al. (2013), Wang, Zhan, Nie, & Xu (2013), Adhau, Mittal, & Mittal (2012), Homberger (2012), Mao et al. (2009), Confessore, Giordani, & Rismondo (2007), Homberger (2007)
Local (L)	Ahmeti & Musliu (2021), Fu & Zhou (2021), He et al. (2021), Rostami & Bagherpour (2020), Tian et al. (2020), Gómez, Fernández, & Castro (2019), Jedrzejowicz & Ratajczak-Ropel (2019), Shi et al. (2019), Villafáñez, Poza, López-Paredes, Pajares, & del Olmo (2019), Li & Xu (2018), Zhang & Chen (2018), Geiger (2017), Asta et al. (2016), Wauters et al. (2016), Wauters, Verbeeck, De Causmaecker, & Berghe (2015), Can & Ulusoy (2014), Zheng et al. (2014), Adhau et al. (2013), Wang et al. (2013), Adhau et al. (2012), Homberger (2012), Confessore et al. (2007), Homberger (2007)
Non-renewable (NR)	Davari Ardakani & Dehghani (2022), Ahmeti & Musliu (2021), Li et al. (2021), Araujo et al. (2020), Cheng et al. (2019), Zhang & Chen (2018), Geiger (2017), Shariatmadari, Nahavandi, Zegordi, & Sobhiyah (2017), Asta et al. (2016), Gholizadeh-Tayyar et al. (2016), Tayyar et al. (2016), Toffolo et al. (2016), Wauters et al. (2016), Beşikci et al. (2015), Can & Ulusoy (2014), Liu, Shan, & Wu (2014), Rokou et al. (2014), Xu & Feng (2014), Tasan & Gen (2013), Krüger & Scholl (2010), Lee & Lei (2001), Chen (1994), Speranza & Vercellis (1993)
Transfer time (TT)	Fu & Zhou (2021), Van Den Eeckhout et al. (2021), Liu & Lu (2019), Suresh et al. (2015), Adhau et al. (2013), Cai & Li (2012), Krüger & Scholl (2010), Krüger & Scholl (2009)
Resource cost (RC)	Li et al. (2021), Van Den Eeckhout et al. (2021), Kannimuthu et al. (2020), Zhang & Chen (2018), Shariatmadari et al. (2017), Gholizadeh-Tayyar et al. (2016), Tayyar et al. (2016), Suresh et al. (2015), Beşikci et al. (2015), Can & Ulusoy (2014), Chen et al. (2014), Liu et al. (2014), Wang et al. (2013), Kolisch & Heimerl (2012), Ning et al. (2012), Mao et al. (2009), Lee & Lei (2001), Kolisch (2000), Chen (1994), Vercellis (1994), Speranza & Vercellis (1993), Kim & Schniederjans (1989)

consequence Van Den Eeckhout, Vanhoucke, & Maenhout (2021), Gholizadeh-Tayyar et al. (2016), Tayyar et al. (2016), and Kolisch & Heimerl (2012) considered the resource rent (RR).

Table 5 shows the existing research in the reviewed literature that addresses resource-based features. Note that only features considered in at least five works are listed, the other cases are referenced in this section.

4.1.4. Relationship-based features

In a job schedule environment, there might be many relationships between jobs. The basic relationship between jobs in the RCMPSP is the end-start relationship, which establishes that a job execution cannot start unless all its predecessor jobs have been fully executed.

Tian et al. (2018), and Pritsker et al. (1969) studied the concurrent or non-concurrent jobs execution (C/N). The concurrency can be established through sets of jobs that must be executed simultaneously, and each job must belong exactly to one set. On the other hand, in the non-concurrency relationship, the sets of jobs represent jobs that cannot be executed in parallel. This relationship might be useful in the case that conflicts between jobs need to be incorporated. In addition, there may be jobs that need to be executed exclusively, e.g., a fumigation job in space requires the stoppage of the rest of the jobs required by the project. This kind of relationship was studied by Hao et al. (2010) under the concept of exclusiveness (EX) and can generate delays in the production line.

The joint consideration of basic relationships (i.e., start-start, start-end, end-start, and end-end) was considered in the generalized variant (GL) in the research of Gholizadeh-Tayyar et al. (2016), Tayyar et al. (2016), and Chen & Shahandashti (2009). Start-start establishes that the execution of a job cannot be started until the execution of the related jobs is started. End-end refers to the fact that a job cannot finish its execution until the related jobs are fully executed. Start-end implies that a job cannot finish its execution until the execution of the related jobs is started. As a feature related, time lags (TL) studied by Davari Ardakani & Dehghani (2022), Joo & Chua (2017), Gholizadeh-Tayyar et al. (2016), and Tayyar et al. (2016) establish a minimum or maximum elapse time between start and completion of the execution of the related jobs.

4.1.5. Project-based features

The simultaneous execution of several projects is one of the main RCMPSP features that increase complexity compared to the single project execution. In the base problem, the simultaneous execution of several projects assumed there are no preferences on any project, and existing projects, respective arrival times to the production line, and due dates are known.

In several environments, preferences or priorities on a project are needed to model daily operations (e.g., a shop that accepts special orders or a hospital emergency room). This feature known as weight (W), is closely related to preemption and can result in a wide range of analyzes, for example, related to costs and time. One of the most studied analyzes is related to penalties incurred when a project is delayed, given the direct proportionality between weight and penalty. This weight can be defined in diverse ways. For example, Nabipoor, Aghaie, & Najafi (2020) establish an explicit degree of weight for each project, and Kannimuthu et al. (2020) assign a different delay penalty for each project. In both cases, this priority or penalty is multiplied by the delay time of each project. In addition, Tian, Liu, & Zhang (2020) assign different priorities for each project related to the use of resources, so in a conflict, resources are assigned to the project with the highest priority. In all cases, the objective is to prioritize the execution of the jobs corresponding to the most critical projects. On the other hand, a service company can receive a new project associated with economic and social improvements, so it could be relevant to accept the project even if it is not planned. This situation captures the uncertainty regarding project arrival (PU) and permits addressing different situations. For example, Satic, Jacko, & Kirkbride (2022) establish that all projects have a probability of arrival, whereas Chen, Ding, Zhang, & Qin (2019) present initial projects with an established arrival date and a set of new arrival projects. In the latter case, at a random period of time, if the number of projects in execution is less than the maximum number allowed, a project is randomly selected from the new arrival set to start its execution. Furthermore, new projects may present different priority levels when compared to existing or already running projects. For example, Wang et al. (2015) establish that the new projects, called urgent projects, have a higher priority regarding using resources. Similar to the job selection concept, project selection (PS) refers to the existence of a set of projects from which to decide the projects to be executed. Existing projects can differ in the number of jobs, execution modes, resource consumption, and costs associated with resources or delays, among other features. The selected projects can be related to deadlines, budget constraints, and maximum or the minimum number of projects that can be selected or executed simultaneously. Tasan & Gen (2013) present the project selection through a graph network containing subgraphs. Each sub-

Table 6
Research works that address project-based features.

Features	Research works
Project selection (PS)	Davari Ardakani & Dehghani (2022), Namazian et al. (2019), Amirian & Sahraeian (2017), Shariatmadari et al. (2017), Namazian & Yakhchali (2016), Shou et al. (2014), Tasan & Gen (2013), Chen & Askin (2009), Gutjahr et al. (2008), Kolisch & Meyer (2006)
Project uncertainty (PU)	Satic et al. (2022), Chen et al. (2019), Hu et al. (2015), Wang et al. (2015), Kao, Hsieh, & Yeh (2006)
Weight (W)	Satic et al. (2022), Kannimuthu et al. (2020), Liu & Xu (2020), Nabipoor et al. (2020), Tian et al. (2020), Cheng et al. (2019), Liu & Lu (2019), Li & Xu (2018), Xin, Su, Wang, & Wang (2018), Zhang & Chen (2018), Beşikci et al. (2015), Wang et al. (2015), Liu et al. (2014), Singh (2014), Xu & Feng (2014), Adhau et al. (2013), Wang et al. (2013), Adhau et al. (2012), Xu & Zhang (2012), Hu, Hu, Huang, Chen, & Song (2010), Mao et al. (2009), Man, Wei, Xiang, & Lishan (2008), Li & Liu (2005), Chiu & Tsai (2002), Lee & Lei (2001), Kolisch (2000), Chen (1994), Vercellis (1994), Speranza & Vercellis (1993), Kim & Schniederjans (1989), Mohanty & Siddiq (1989)

Table 7
Research works that address project cost/profit (PCP) feature.

Davari Ardakani & Dehghani (2022), Namazian et al. (2019), Amirian & Sahraeian (2017), Shariatmadari et al. (2017), Namazian & Yakhchali (2016), Suresh et al. (2015), Can & Ulusoy (2014), Shou et al. (2014), Chen & Askin (2009), Gutjahr et al. (2008), Chiu & Tsai (2002), Chen (1994), Vercellis (1994), Speranza & Vercellis (1993), Kim & Schniederjans (1989)

graph comprises a set of alternative projects, establishing disjunctive (OR relations) or conjunctive (AND relations) relationships.

Table 6 shows the existing research in the reviewed literature that addresses project-based features. Note that only features considered in at least five works are listed, the other cases are referenced in this section.

4.1.6. Project- and relationship-based features

The precedence relationships have been mainly studied between jobs belonging to the same project. However, Ning, Dongdong, Martin, & Zuhair (2012) studied relationships between projects, i.e., zone and blocked area. The concurrency relationship mentioned above in Section 4.1.4 can be extended to jobs from different projects under the concept of zone (Z). A set of jobs belonging to different projects must be executed concurrently, maintaining the same duration but executing a common resource consumption. For example, pieces from different projects must be cooled, and a single freezer is used. On the other hand, in a period, a specific reference project can operate as a point of reference for transportation of the resources required by neighboring projects. This set of projects constitutes a blocked area (B), which cannot be accessed during this period. Consequently, executing any job of any project in the blocked area is impossible.

4.1.7. Project- and resource-based features

Similar to job cost/profit, the project cost/profit (PCP) is studied in a multi-project environment with budget considerations. Chen (1994) established an average maintenance cost for each project in each time period. That cost together with others (e.g., the cost of using resources) cannot exceed the available budget in each time period. The completion of a project can also represent an inflow of cash that may (or may not) depend on the time period at which the project ends. For example, Suresh et al. (2015) establish that the obtained benefit is independent of the time period in which each project ends, but it directly influences the cash flow. Whereas Shou, Xiang, Li, & Yao (2014) establish a different profit for the completion of the project at each time period. On the other hand, Can & Ulusoy (2014) simultaneously consider that the start and completion of a project represent an outflow and inflow of cash. The study of this feature is closely related to the selection of projects, and in the works where it is considered, maximizing the net present value is the most frequently pursued objective function.

Table 7 shows the existing research in the reviewed literature that addresses the project cost/profit feature.

4.1.8. Research on the problem features and time perspective

In the collected literature in this survey, 82% of the works consider the base problem linked to different features. There are significant differences between the number of works that study resource- and relationship-based features 52% and 6%, respectively. In the study of resource-based features, the existence of local and non-renewable resources are the most representative, with 20% each. Concerning research addressing projects- and jobs-based features, the difference between the usage percentages is 19%, being projects-based features the most significant with 39%. The most studied projects- and jobs-based features are weight and uncertainty with 27% and 12%, respectively.

The problem evolution is depicted in Fig. 6. The discussed features are associated with the year in which they were presented. The bars indicate the amount of research carried out from that moment, and the colors follow the pattern used in Fig. 5.

To analyze relevant research topics related to the problem in recent years, Fig. 7 shows, since 2010, how the research was carried out concerning problem features studied in at least ten research works.

The study of new features associated with the problem exhibited a slow growth until the beginning of this century. Since 2010, its growth has increased, resulting in more than 34% of the features associated with the problem. This may be related to the increase in research carried out since 2010. Note that there are features that, after being presented, were not considered again, some were recently presented, and others are associated with features less studied in the literature, as shown in Fig. 6.

Figures 6 and 7 show that the most representative features are the use of local and non-renewable resources, the existence of weights associated with projects, multiple execution modes, resource costs, decentralized resource allocation, and job duration uncertainty. In many cases, the combination of local resources and decentralized resource allocation is addressed. This finds its rationale bearing in mind the relationship between these features and the rise of decentralized production models. In the same way, the study of non-renewable resources is closely related to multiple execution modes. The use of non-renewable resources was added to the problem in 1993, presenting 23 related research works, 87% of them in the last decade, representing one of the features with the most significant increase in related research in this period. The existence of weight associated with projects is the most reflected feature in the literature, with notable growth since 2010, presenting 65% of related research since this date. Regarding the uncertainty, this feature generates a higher degree of complexity to the problem but is necessary to study given its impact on the schedule generation. The study of this feature related to job duration increased,

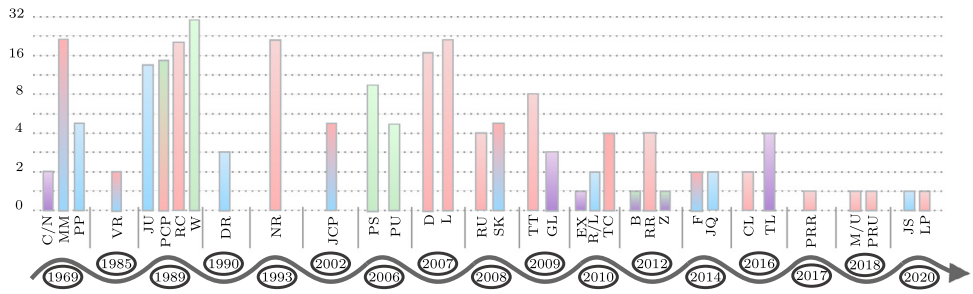

Fig. 6. Evolution of the problem over the years.

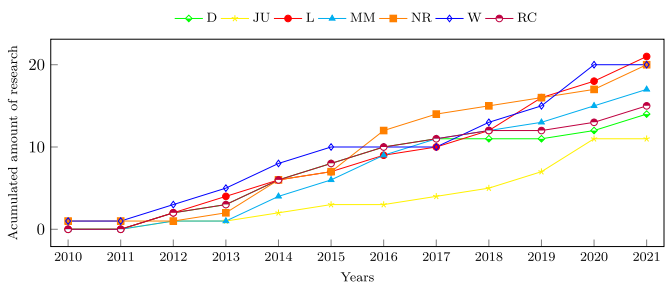

Fig. 7. Main problem features studied since 2010.

presenting 79% of investigations in the 2010–2021 period, with a significant increase of 57% in the last five years. A large number of research works can also be observed that addressed cost considerations, whether related to resources, jobs, or projects. These features bring the problem closer to real environments where fundamental elements are costs/profit and budgets. It is important to highlight that, as seen in the figures, researching the base problem is still relevant, despite having been presented more than 50 years ago.

4.2. Objective functions

The objective functions are classified based on three main factors: resources, projects, and time. Figure 8 shows the classification scheme proposed. The colors red, green, and orange represent objective functions based on resources, projects, and time, respectively.

4.2.1. Resource-based objective functions

The previous section discussed several resource-based features. They led to the emergence of several objective functions aimed at efficient resource use and cost evaluation.

- *Economic*: The valuation of economic aspects is fundamental in project scheduling environments. When assessing discontinued cash flow caused by cost-related features, it may be necessary to control cash stock to avoid situations where the production process is stopped due to a lack of cash. Because of this, the net present value maximization (NPV) was proposed. As a particular case, Zapata et al. (2008) studied the maximization of the expected net present value by evaluating uncertainty. Regarding evaluating resource costs, the total project cost minimization (TPC), also called the sum of costs, considers the costs related to the use, transfer, inactivity, rent, reserving, and maintenance of the resources. In particular cases, Chen (1994) addressed the cost of the critical project where only the critical projects are valued, Kolisch & Heimerl (2012) studied the labor costs influenced by different skills and efficiency of human personnel, and Rostami & Bagherpour (2020) addressed the sum of construction costs related to resource facility location. Finally, obtaining profits by completing a job and/or a project resulted in the objective of maximizing the total project profit (TPP) that can be evaluated at the end of one or all projects. This objective function is closely related to job and/or project selection features.
- *Consumption*: Resource inactivity occurs when resources are available but not required by any job; this can be due to precedence constraints. This aspect has been studied under the same conditions through two objective functions, i.e., the total resource idle time minimization and the idle resources minimization (IR). Since both measures address the same objective, we refer to this objective function as idle resource minimization. When studying resources in project scheduling, it is essential to assess factors associated with the resource inventory. For instance, resource inventory limitations can cause the execution of jobs corresponding to a project not to be consecutive. This results in additional expenses due to projects waiting for the execution of the related jobs being part of the inventory. Hence, it is necessary to minimize the number of in-process inventory jobs (IPI), an objective function studied by Lova, Maroto, & Tormos (2000) and Kim & Schniederjans (1989). On the other hand, stable resource consumption is one of the priorities present in the objective functions that seek to avoid peaks in resource usage. This objective was initially named resource average load factor minimization. However, at the beginning of this century, its study began under the name of resource-leveling maximization (RL), a term used in this work.

Table 8 shows the existing research addressing resource-based objective functions. Note that only the objective functions considered in at least five works are listed, the other cases are referenced in this section.

4.2.2. Time-based objective functions

The following classification groups those objective functions that evaluate the completion time or the delays that may occur in the completion of projects.

- *Delays*: Project delays occur when the completion time is larger than the due date. The project delay is defined as the difference between the makespan and the desired due date if the makespan is greater than the desired due date; otherwise, the delay is zero. There are several objective functions associated with the delays. The average project delay minimization (APD) is the most used objective function regarding this aspect. Considering that the number of projects is a parameter of the problem, in this review, the objective functions based on the total delays are classified as average project delay (e.g., total project delay). The standard deviation allows for evaluating the dispersion of a data set, so Dumond (1992) and Dumond & Mabert (1988) studied as an objective the standard deviation mean lateness minimization (SDML) to balance the delays incurred in each project. Other measures based on delays and efficiency of the generated schedules were studied under average percent project delay, efficiency, and robustness. In this review, the measures related to efficiency, either in average or total values, are classified as minimization of

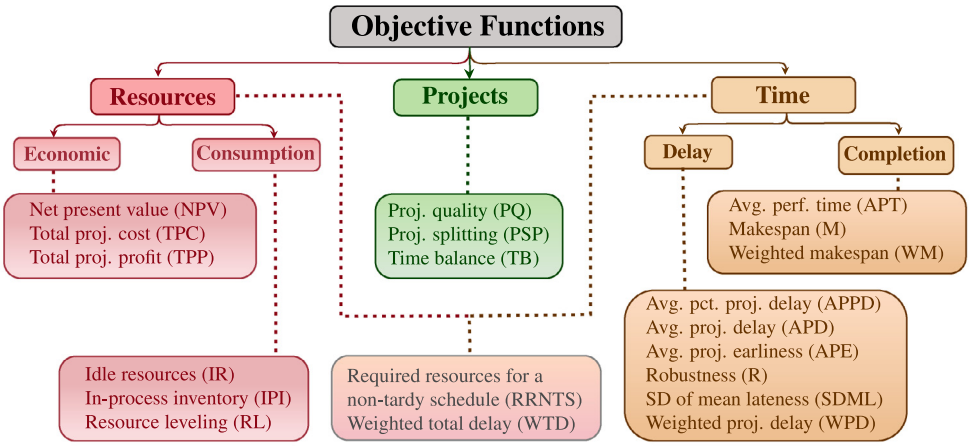

Fig. 8. Classification of the RCMPSP objective functions.

Table 8
Research works that address resource-based objective function.

Objective function	Research works
Economics	
Net present value (NPV)	Davari Ardakani & Dehghani (2022), He et al. (2021), Suresh et al. (2015), Can & Ulusoy (2014), Zapata et al. (2008), Chiu & Tsai (2002), Vercellis (1994), Speranza & Vercellis (1993)
Total proj. profit (TPP)	Satic et al. (2022), Namazian et al. (2019), Amirian & Sahraeian (2017), Shariatmadari et al. (2017), Shou et al. (2014), Wang et al. (2013), Chen & Askin (2009), Gutjahr et al. (2008), Kolisch & Meyer (2006)
Total proj. cost (TPC)	Fu & Zhou (2021), Van Den Eeckhout et al. (2021), Kannimuthu et al. (2020), Rostami & Bagherpour (2020), Cheng et al. (2019), Liu & Lu (2019), Zhang & Chen (2018), Amirian & Sahraeian (2017), Gholizadeh-Tayyar et al. (2016), Tayyar et al. (2016), Chen et al. (2014), Liu et al. (2014), Singh (2014), Xu & Feng (2014), Adhau et al. (2013), Wang et al. (2013), Kolisch & Heimerl (2012), Ning et al. (2012), Krüger & Scholl (2010), Mao et al. (2009), Chen (1994)
Consumption	
Idle resources (IR)	Hauder et al. (2020), Shi et al. (2019), Amirian & Sahraeian (2017), Krüger & Scholl (2010), Lova et al. (2000), Mohanty & Siddiq (1989)
Resource leveling (RL)	Davari Ardakani & Dehghani (2022), Issa et al. (2021), Liu & Lu (2019), Mao et al. (2009), Lova et al. (2000), Tsubakitani & Deckro (1990), Kim & Schniederjans (1989)

the average percent project delay (APPD) if the job duration is deterministic; and maximization of robustness (R) in the cases of the research presented by Zhu et al. (2021), Chen et al. (2019), and Wang, He, Kerkhove, & Vanhoucke (2017). On the other hand, different weights for each project greatly influence schedule generation. Weights can be associated with penalties for delays; the higher the weight, the higher the delay penalty. Given that, the weighted project delay minimization (WPD) arises as an objective function. Gonçalves, de Magalhães Mendes, & Resende (2015), Krüger & Scholl (2010), Gonçalves, Mendes, & Resende (2008), and Bock & Patterson (1990) pursue the average project earliness minimization (APE) considering that the early completion sometimes negatively influences the project delivery, so it can also be penalized with the same or a different value used for delays.

- *Completion*: The project completion time or makespan (M) is defined as the time when all jobs associated with a project are fully executed. Performance time, also known as flow time or throughput time, is defined as the difference between the first job execution start time of a project and a project makespan. In a single project environment, the performance time minimization (APT) is equivalent to the makespan minimization. This situation does not happen in a multi-project environment where the makespan is defined as the maximum makespan among all projects and the average performance time as the average performance times of all projects. In this sense, the average performance time minimization attempts to conduct projects as soon as possible. Since the number of projects is a fixed value, minimizing the total performance time is classified as the minimization of the average performance time. Weights are used to minimize the weighted makespan (WM) where the makespan value is multiplied by the corresponding weight, so the higher the weight, the greater the influence on the objective function.

Table 9 shows the existing research in the reviewed literature that addresses the time-based objective functions. Note that only the objective functions considered in at least five works are listed, the other cases are referenced in this section.

4.2.3. Resource- and time-based objective functions

Project completion time and resource use are two of the most studied factors in the literature. In the previous sections, the related objective functions were defined individually. Nevertheless, in the sequel, we discuss the joint consideration of both objective types.

The weight or cost associated with resources motivated the weighted total delay minimization (WTD) objective. That objective function was used by Mohanty & Siddiq (1989) to minimize the sum of total resources demanded by each project multiplied by the corresponding weighting and delay. Studying multiple execution modes or job selection can result in schedules with variable resource use. In this sense, Lee & Lei (2001) studied the resources use, prioritizing on-time project completion, an objective known as the minimization of required resources for a non-tardy schedule (RRNTS).

4.2.4. Project-based objective functions

In Section 4.1, it can be seen that the project-based features were not extensively studied, whereby there is not a wide range of objective functions that contain project-related aspects.

In the context of continuous project execution, Lova et al. (2000) propose the project splitting minimization (PSP) with the

Table 9
Research works that address time-based objective function.

Objective function	Research works
Delay	
Avg. pct. proj. delay (APPD)	Van Eynde & Vanhoucke (2020), Chen et al. (2019), Chakrabortty et al. (2017), Wang et al. (2017), Pérez et al. (2016), Vázquez et al. (2015), Adhau et al. (2013), Cai & Li (2012), Browning & Yassine (2010b), Krüger & Scholl (2009), Lova & Tormos (2001), Mohanty & Siddiq (1989)
Avg. proj. delay (APD)	Ahmeti & Musliu (2021), Araujo et al. (2020), Gómez et al. (2019), Jedrzejowicz & Ratajczak-Ropel (2019), Shi et al. (2019), Chakrabortty et al. (2017), Geiger (2017), Joo & Chua (2017), Asta et al. (2016), Toffolo et al. (2016), Wauters et al. (2016), Gonçalves et al. (2015), Wauters et al. (2015), Singh (2014), Zheng et al. (2014), Adhau et al. (2012), Homberger (2012), Browning & Yassine (2010b), Krüger & Scholl (2009), Gonçalves et al. (2008), Lova & Tormos (2001), Tsai & Chiu (1996), Dumond (1992), Bock & Patterson (1990), Dumond & Mabert (1988), Mohanty & Siddiq (1989)
Weighted proj. delay (WPD)	Li et al. (2021), Liu & Xu (2020), Nabipoor et al. (2020), Tian et al. (2020), Cheng et al. (2019), Li & Xu (2018), Zhu et al. (2018), Beşikci et al. (2015), Wang et al. (2015), Tasan & Gen (2013), Ju & Chen (2012), Xu & Zhang (2012), Krüger & Scholl (2010), Lee & Lei (2001), Kolisch (2000), Lawrence & Morton (1993), Kim & Schniederjans (1989), Pritsker et al. (1969)
Completion	
Avg. perf. time (APT)	Gonçalves et al. (2015), Chen et al. (2014), Gonçalves et al. (2008), Dumond (1992), Deckro et al. (1991), Bock & Patterson (1990), Dumond & Mabert (1988), Pritsker et al. (1969)
Makespan (M)	Davari Ardakani & Dehghani (2022), Satic et al. (2022), Ahmeti & Musliu (2021), Cui et al. (2021), Issa et al. (2021), Li et al. (2021), Zhu et al. (2021), Hauder et al. (2020), Kannimuthu et al. (2020), Tian et al. (2020), Gómez et al. (2019), Jedrzejowicz & Ratajczak-Ropel (2019), Namazian et al. (2019), Villafáñez et al. (2019), Song et al. (2018), Tian et al. (2018), Zhang & Chen (2018), Geiger (2017), Asta et al. (2016), Namazian & Yakhchali (2016), Pérez et al. (2016), Toffolo et al. (2016), Wauters et al. (2016), Chen & Ju (2015), Hu et al. (2015), Sonmez & Uysal (2015), Vázquez et al. (2015), Wauters et al. (2015), Can & Ulusoy (2014), Chen et al. (2014), Rokou et al. (2014), Singh (2014), Tasan & Gen (2013), Homberger (2012), Ju & Chen (2012), Ning et al. (2012), Xu & Zhang (2012), Chen & Shahandashti (2009), Confessore et al. (2007), Homberger (2007), Linyi & Yan (2007), Kumanan et al. (2006), Kao et al. (2006), Zhuang & Yassine (2004), Lee & Lei (2001), Chen (1994), Speranza & Vercellis (1993), Pritsker et al. (1969)
Weighted makespan (WM)	Xin et al. (2018), Zhu et al. (2018), Xu & Feng (2014), Hu et al. (2010), Man et al. (2008), Li & Liu (2005)

aim of avoiding additional efforts related to project control. The incorporation of multiple execution modes, jobs selection, and/or quality resulted in several objective functions, such as the project quality maximization (PQ), given by the quality of the average jobs, studied by Kannimuthu et al. (2020) and Xu & Feng (2014). Also influenced by the mentioned features, Hauder et al. (2020) studied the time balance maximization (TB) pursuing the lowest difference between the minimum and maximum duration of the selected jobs.

4.2.5. Multiple objectives

The wide range of RCMPSP-related features causes that, in many studies, the problem is solved considering several objectives. Note that this does not necessarily imply multi-objective functions approaches but to propose more than one independently objective for a given variant.

In the collected literature, many works pursued different objective functions analyzing a trade-off, for example, Issa et al. (2021), Hauder et al. (2020), Cheng, Lo, & Chu (2019), Jedrzejowicz & Ratajczak-Ropel (2019) and Chakrabortty, Sarker, & Essam (2017). In contrast, several studies presented multi-objective functions, and some divided the problem into phases and pursued different objective functions in each one, for example, Davari Ardakani & Dehghani (2022), Kannimuthu et al. (2020), Namazian, Haji Yakhchali, & Rabbani (2019), Amirian & Sahraeian (2017) and Asta et al. (2016).

4.2.6. Research on the objective functions and analysis

Figure 9 (a) shows the works related percentage to the different classifications proposed. For a more in-depth review, Fig. 9(b) and (c) show the percentage of works that consider time-based and resource-based objective functions, respectively. The time-based objective functions studied in less than ten research works are grouped in the category *Others*.

As can be observed, the time-based objective functions are the most studied, followed by resource-based. On the other hand, project-based objective functions represent a small portion. The most studied objective functions are the makespan minimization and the average project delay minimization, corresponding to completion and delay sub-classifications. The total project cost minimization is the next most pursued objective, followed by the minimization of the weighted project delay and the average percentage of project delay.

The first objective function in the multi-project environment presented in 1969 by Pritsker et al. (1969) was related to time optimization. Despite the elapsed years until today, and although the most studied features are related to resources, the percentages distribution indicates that time optimization is the main factor in the problems related to the RCMPSP.

5. Solution methods

In the beginning, the RCMPSP-related variants were solved mainly through mathematical models and priority rule-based heuristics. However, given problem complexity and the emergence of new solution techniques, the range of proposed solution methodologies was expanded.

Figure 10 shows the solution methods classification applied to the RCMPSP and related variants, including exact, approximate, and hybrid algorithms. The yellow and blue colors represent solution methods related to exact and approximate algorithms. The hybrid algorithms based only on approximate algorithms are represented in blue, and the hybrid algorithms supported in both categories (exact and approximate algorithms) are represented by combining both colors. Given the number of different solution methods, similar algorithms used in less than five research works are clustered. Namely, algorithms based on Lagrangian relaxation, goal programming, branch and bound, column generation, and the resolution of the mathematical model are clustered as *Linear Programming-based*. *Swarm Intelligence* contains the algorithms based on ant colony optimization, particle swarm optimization, and artificial bee colony. The research works based on memetic algorithms are clustered as genetic algorithms. Lastly, research works based on variable neighborhood search, critical chain method, robust optimization, gravitational search algorithm, learning, and specific local search heuristics are grouped in the *Other algorithms* category.

Figure 11 (a) and (b) show the use percentages of the approximate and exact algorithms, respectively.

Tables 10 and 11 show the works that used each approximate or exact algorithm to solve the problem, respectively. Note the pri-

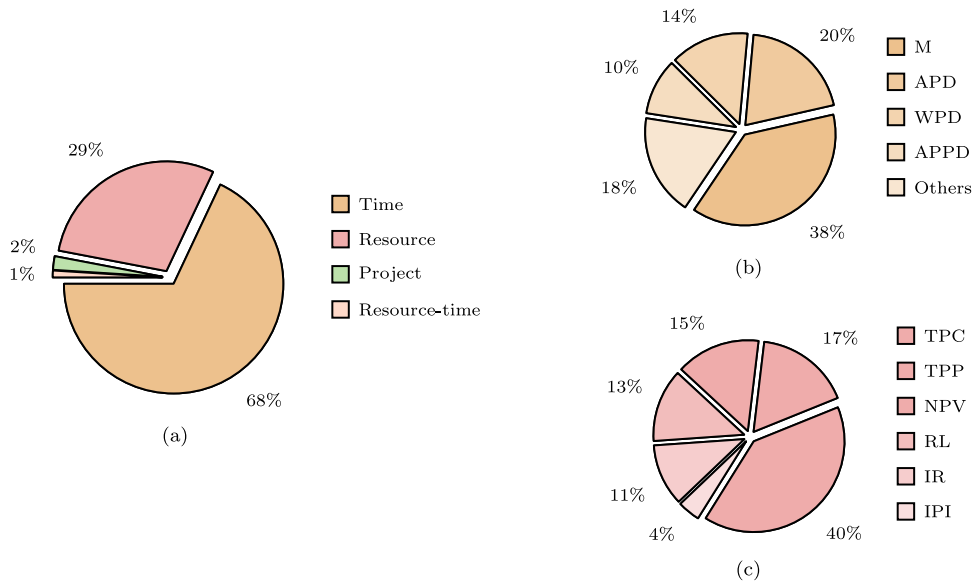

Fig. 9. Percentage of research by classification of objective function studied.

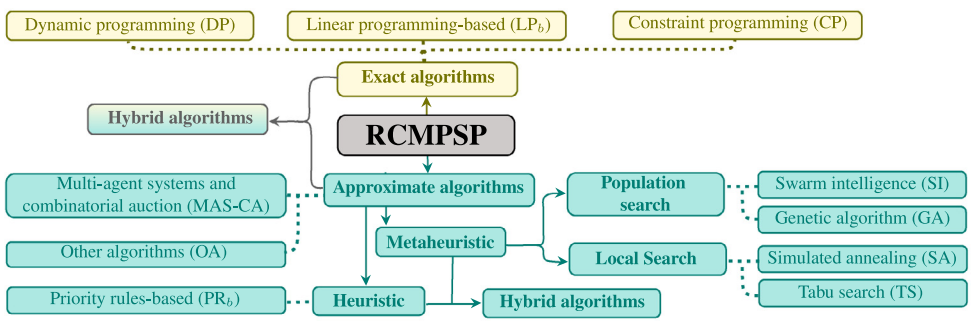

Fig. 10. Classification of algorithms to solve the RCMPSP and existing variants.

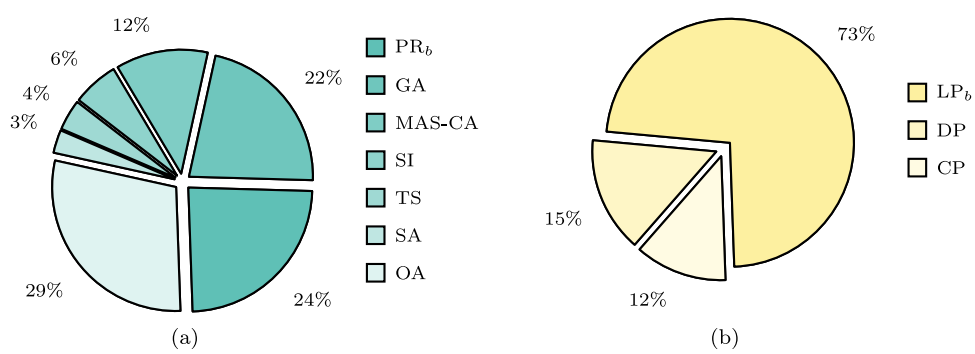

Fig. 11. Percentage of different solution methods applied to the RCMPSP.

ority rule-based group include only defined heuristics. In the collected works, the implicit use of priority rules was found in the design of the algorithms used.

To identify which are the main solution methods for solving the problem in recent years, Fig. 12 shows, since 2010, the research carried out for the methods that have been studied in at least ten research works.

The approximate and hybrid algorithms are the most used in the literature to solve the RCMPSP; this can be due to the high problem complexity level. Linear programming-based algorithms are the most used exact algorithms. At the same time, heuristics based on priority rules and genetic algorithms are the most repre-

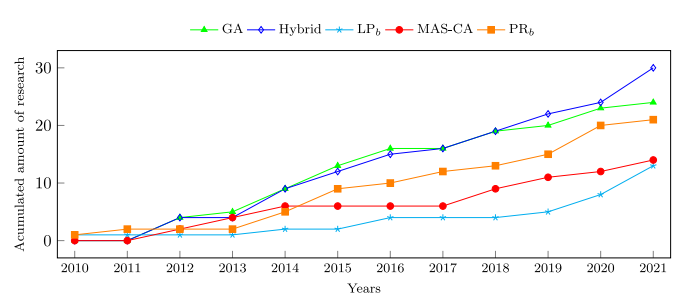

Fig. 12. Main solution methods applied since 2010.

Table 10
Reports of approximate solution methods for RCMPSP variants.

Solution methods	Research works
Genetic algorithm (GA)	Satic et al. (2022), Fu & Zhou (2021), Rostami & Bagherpour (2020), Van Eynde & Vanhoucke (2020), Namazian et al. (2019), Li & Xu (2018), Tian et al. (2018), Xin et al. (2018), Asta et al. (2016), Pérez et al. (2016), Tayyar et al. (2016), Beşikci et al. (2015), Gonçalves et al. (2015), Sonmez & Uysal (2015), Suresh et al. (2015), Can & Ulusoy (2014), Rokou et al. (2014), Shou et al. (2014), Zheng et al. (2014), Tasan & Gen (2013), Cai & Li (2012), Kolisch & Heimerl (2012), Ning et al. (2012), Xu & Zhang (2012), Chen & Shahandashti (2009), Gonçalves et al. (2008), Gutjahr et al. (2008), Man et al. (2008), Kumanan et al. (2006), Li & Liu (2005), Zhuang & Yassine (2004)
Hybrid algorithm	Ahmeti & Musliu (2021), Fu & Zhou (2021), He et al. (2021), Issa et al. (2021), Van Den Eeckhout et al. (2021), Zhu et al. (2021), Liu & Xu (2020), Tian et al. (2020), Gómez et al. (2019), Liu & Lu (2019), Namazian et al. (2019), Li & Xu (2018), Tian et al. (2018), Zhu et al. (2018), Chakrabortty et al. (2017), Asta et al. (2016), Toffolo et al. (2016), Sonmez & Uysal (2015), Vázquez et al. (2015), Wang et al. (2015), Can & Ulusoy (2014), Rokou et al. (2014), Shou et al. (2014), Xu & Feng (2014), Zheng et al. (2014), Cai & Li (2012), Kolisch & Heimerl (2012), Ning et al. (2012), Xu & Zhang (2012), Chen & Shahandashti (2009), Gutjahr et al. (2008), Man et al. (2008), Confessore et al. (2007), Chen (1994)
Multi-agent systems and combinatorial auction (MAS-CA)	Fu & Zhou (2021), Li et al. (2021), Liu & Xu (2020), Cheng et al. (2019), Jedrzejowicz & Ratajczak-Ropel (2019), Li & Xu (2018), Song et al. (2018), Zhang & Chen (2018), Shou et al. (2014), Zheng et al. (2014), Adhau et al. (2013), Wang et al. (2013), Adhau et al. (2012), Homberger (2012), Mao et al. (2009), Confessore et al. (2007), Homberger (2007)
Other algorithms (OA)	Satic et al. (2022), Ahmeti & Musliu (2021), Cui et al. (2021), Van Den Eeckhout et al. (2021), Zhu et al. (2021), Kannimuthu et al. (2020), Nabipoor et al. (2020), Rostami & Bagherpour (2020), Tian et al. (2020), Gómez et al. (2019), Liu & Lu (2019), Villafáñez et al. (2019), Li & Xu (2018), Zhu et al. (2018), Amirian & Sahraeian (2017), Chakrabortty et al. (2017), Geiger (2017), Shariatmadari et al. (2017), Asta et al. (2016), Namazian & Yakhchali (2016), Toffolo et al. (2016), Hu et al. (2015), Sonmez & Uysal (2015), Vázquez et al. (2015), Wauters et al. (2015), Zheng et al. (2014), Cai & Li (2012), Ju & Chen (2012), Ning et al. (2012), Xu & Zhang (2012), Hao et al. (2010), Hu et al. (2010), Chen & Askin (2009), Krüger & Scholl (2009), Kao et al. (2006), Chiu & Tsai (2002), Lee & Lei (2001), Speranza & Vercellis (1993), Tsubakitani & Deckro (1990), Kim & Schniederjans (1989)
Priority rules-based (PR_b)	Satic et al. (2022), Issa et al. (2021), Kannimuthu et al. (2020), Liu & Xu (2020), Rostami & Bagherpour (2020), Van Eynde & Vanhoucke (2020), Chen et al. (2019), Shi et al. (2019), Tian et al. (2018), Chakrabortty et al. (2017), Wang et al. (2017), Pérez et al. (2016), Chen & Ju (2015), Suresh et al. (2015), Vázquez et al. (2015), Wang et al. (2015), Shou et al. (2014), Singh (2014), Xu & Feng (2014), Zhang & Sun (2011), Browning & Yassine (2010b), Kolisch & Meyer (2006), Lova & Tormos (2001), Kolisch (2000), Tsai & Chiu (1996), Chen (1994), Lawrence & Morton (1993), Dumond (1992), Bock & Patterson (1990), Mohanty & Siddiq (1989), Dumond & Mabert (1988), Kurtulus (1985), Kurtulus & Davis (1982), Pritsker et al. (1969)
Simulated annealing (SA)	He et al. (2021), Joo & Chua (2017), Sonmez & Uysal (2015), Chen & Shahandashti (2009), Man et al. (2008)
Swarm intelligence (SI)	Gómez et al. (2019), Tian et al. (2018), Chen & Ju (2015), Liu et al. (2014), Rokou et al. (2014), Xu & Feng (2014), Gutjahr et al. (2008), Linyi & Yan (2007)
Tabu search (TS)	Ahmeti & Musliu (2021), He et al. (2021), Zhu et al. (2021), Tian et al. (2018), Zhu et al. (2018), Kolisch & Heimerl (2012)

Table 11
Reports of exact solution methods for RCMPSP variants.

Solution methods	Research works
Constraint programming (CP)	Ahmeti & Musliu (2021), Hauder et al. (2020), Liu & Lu (2019)
Dynamic programming (DP)	Satic et al. (2022), Wang et al. (2015), Chen et al. (2014), Confessore et al. (2007)
Linear programming-based (LP_b)	Davari Ardakani & Dehghani (2022), Cui et al. (2021), Issa et al. (2021), Li et al. (2021), Van Den Eeckhout et al. (2021), Araujo et al. (2020), Hauder et al. (2020), Rostami & Bagherpour (2020), Namazian et al. (2019), Gholizadeh-Tayyar et al. (2016), Toffolo et al. (2016), Can & Ulusoy (2014), Krüger & Scholl (2010), Zapata et al. (2008), Chen (1994), Vercellis (1994), Speranza & Vercellis (1993), Deckro et al. (1991), Mohanty & Siddiq (1989)

sentative approximate algorithms, with 24% and 22%, respectively. Algorithms based on multi-agent systems and combinatorial auctions are used in 12% of the works. In contrast, algorithms based on swarm intelligence, tabu search, and simulated annealing are found in at most 6% of the works.

During the last decade, there was a relevant increase in the hybrid algorithms applied to the RCMPSP and related variants. To the best of our knowledge, the most used algorithms for hybridization are genetic algorithms, priority rule-based, linear programming-based, tabu search, and multi-agent systems and combinatorial auctions. Priority rule-based heuristics and genetic algorithms are the most used algorithms for solving the problem since the beginning, presenting a stable growth since 2010 and being the most used in the last five years. Moreover, the use of multi-agent systems and combinatorial auctions showed significant growth in the last five years; this can be a consequence of studying decentralized management and local resources. Finally, algorithms based on linear programming showed an important growth in the last two years.

6. Benchmarks

At the beginning of studying the RCMPSP, there were no instance sets to evaluate the performance of the proposed algorithms and most works were validated using problem examples. This practice is still used in recent research, mainly due to the large number of features proposed and the lack of instance sets for each resulting problem variant. In addition, several works evaluate the models and algorithms through own generated instances for the RCMPSP variant addressed.

Kolisch & Sprecher (1997) proposed a project scheduling problem library (PSPLib)[2] for the study of the RCPSP. After it, many works on the RCMPSP used that benchmark in their numerical experiments by combining two or more single-project instances, adding the necessary aspects to cover the studied features, and generating their specific multi-project instances. Moreover, several works were carried out on real instances, demonstrating the importance of this problem within the industry sector. The first and most recognized set of instances is the multi-project scheduling problem library (MPSPLib)[3], presented by Homberger (2007) specifically for decentralized RCMPSP with local resources. MPSPLib contains 80 generated instances from the combination of 2, 5, 10, or 20 problems (projects) of the PSPLib composed of 30, 90, or 120 jobs, assigning an arrival time to each project and al-

[2] http://www.om-db.wi.tum.de/psplib/main.html
[3] http://www.mpsplib.com

Table 12
Reports of research works on different types of instances.

Instances	Research works
Generated instances	Davari Ardakani & Dehghani (2022), Satic et al. (2022), Issa et al. (2021), Li et al. (2021), Van Den Eeckhout et al. (2021), Zhu et al. (2021), Araujo et al. (2020), Hauder et al. (2020), Kannimuthu et al. (2020), Nabipoor et al. (2020), Rostami & Bagherpour (2020), Tian et al. (2020), Van Eynde & Vanhoucke (2020), Chen et al. (2019), Cheng et al. (2019), Liu & Lu (2019), Namazian et al. (2019), Zhu et al. (2018), Amirian & Sahraeian (2017), Chakrabortty et al. (2017), Shariatmadari et al. (2017), Wang et al. (2017), Pérez et al. (2016), Beşikci et al. (2015), Chen & Ju (2015), Gonçalves et al. (2015), Hu et al. (2015), Sonmez & Uysal (2015), Suresh et al. (2015), Wang et al. (2015), Can & Ulusoy (2014), Rokou et al. (2014), Shou et al. (2014), Tasan & Gen (2013), Cai & Li (2012), Ju & Chen (2012), Kolisch & Heimerl (2012), Krüger & Scholl (2010), Hao et al. (2010), Mao et al. (2009), Chen & Askin (2009), Krüger & Scholl (2009), Gonçalves et al. (2008), Linyi & Yan (2007), Confessore et al. (2007), Kolisch & Meyer (2006), Chiu & Tsai (2002), Lova & Tormos (2001), Kolisch (2000), Lova et al. (2000), Tsai & Chiu (1996), Vercellis (1994), Speranza & Vercellis (1993), Lawrence & Morton (1993), Dumond (1992), Tsubakitani & Deckro (1990), Dumond & Mabert (1988), Kurtulus (1985), Kurtulus & Davis (1982)
Problem examples	Shi et al. (2019), Namazian & Yakhchali (2016), Chen et al. (2014), Liu et al. (2014), Singh (2014), Ning et al. (2012), Zhang & Sun (2011), Hu et al. (2010), Man et al. (2008), Zapata et al. (2008), Kumanan et al. (2006), Kao et al. (2006), Li & Liu (2005), Zhuang & Yassine (2004), Lee & Lei (2001), Deckro et al. (1991), Bock & Patterson (1990), Kim & Schniederjans (1989), Mohanty & Siddiq (1989), Pritsker et al. (1969)
Real-world instances	Cui et al. (2021), Kannimuthu et al. (2020), Tian et al. (2020), Song et al. (2018), Tian et al. (2018), Xin et al. (2018), Zhang & Chen (2018), Joo & Chua (2017), Gholizadeh-Tayyar et al. (2016), Tayyar et al. (2016), Sonmez & Uysal (2015), Xu & Feng (2014), Xu & Zhang (2012), Chen & Shahandashti (2009), Gutjahr et al. (2008), Kolisch & Meyer (2006), Chen (1994), Vercellis (1994), Speranza & Vercellis (1993)
MPSPLib	Fu & Zhou (2021), Liu & Xu (2020), Gómez et al. (2019), Jedrzejowicz & Ratajczak-Ropel (2019), Villafáñez et al. (2019), Li & Xu (2018), Pérez et al. (2016), Wauters et al. (2015), Zheng et al. (2014), Adhau et al. (2013), Adhau et al. (2012), Homberger (2012), Homberger (2007)
MISTA-2013	Ahmeti & Musliu (2021), Geiger (2017), Asta et al. (2016), Toffolo et al. (2016), Wauters et al. (2016)

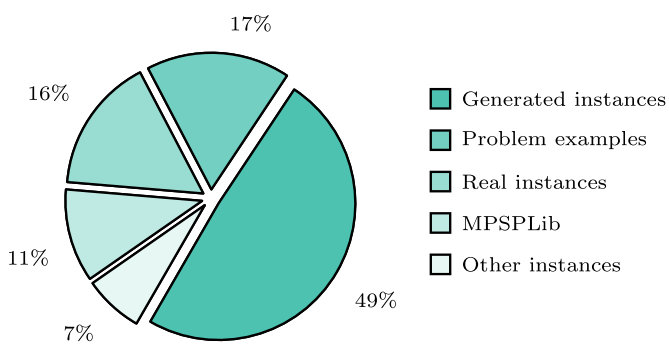

Fig. 13. Percentage of research on a different set of instances.

lowing the use of resources globally and locally. Subsequently, this library was expanded by Homberger (2012) adding 60 new instances that present differences in terms of project access to local resources. MPSPLib allows the online evaluation of proposed solutions and provides various objective functions. Browning & Yassine (2010a) present the first multi-network problem generator (MNPG)[4], also used by He, He, & Wang (2021). The authors generated 12,320 random instances of the problem, but the best values obtained for each possible instance are unknown. Using instances of the MPSPLib and MNPG, a benchmark set known as the resource-constrained multi-project scheduling problem library (RCMPSPLIB)[5] was created by Pérez and Posada and used by Pérez, Posada, & Lorenzana (2016) and Vázquez, Calvo, & Ordóñez (2015). Another variant that has a well-defined set of instances is the multi-mode RCMPSP. This set was developed for the MISTA-2013[6] challenge and is composed of project groups taken from the PSPLib by adding multiple execution modes.

Table 12 shows the research in the reviewed literature that uses each instance set. Note that only the instance sets present in at least five works are listed, and some investigations perform validation with several types of instances.

Figure 13 shows the percentages of works using the different instance sets described above. In the category *Other instances*, the works that used the MISTA-2013, MNPG, and RCMPSPLIB instance sets were grouped.

The highest percentage is represented by the generated sets in the different works. It is important to note that 12% of the research was conducted using real-world based instances. Regarding the instance sets presented in the literature, the MPSPLib is the most used benchmark. The increased use of this library may be due to the frequent study of decentralized conditions and local resource use.

7. Connection to practice

This section analyzes those works that solved real-world-based cases with the aim of assessing the gap between theory and practice concerning the RCMPSP and related variants.

Table 13 shows the research developed in real environments. Column 2 shows the application domain. Columns 3–5 show dimensions of the scenarios handled by each case in terms of the maximum number of projects, the maximum number of jobs, and the maximum number of different renewable resources considered. Following the categorization provided in Section 4, columns 6–13 and 14–34 report the objective function and features, respectively. Moreover, to analyze the different objective functions and features applied in real environments, Fig. 14 shows the percentage of works for (a) objective functions and (b) problem features.

The applications in the construction field are the most representative ones. The maximum number of projects per instance and jobs per project in most works do not exceed 18 and 46, respectively. These values, especially the number of jobs per project, are smaller than the similar benchmark values proposed in the literature. Note there are works, for example, Kolisch & Meyer (2006), that generated a set of instances based on the data collected in the company where the research was conducted. On the other hand, in several works, the number of different renewable global resources exceeds the values found in the generated instance sets.

The most used objectives are related to the time category, followed by resource-based objective functions. Regarding the described features, features associated with jobs-resources, resources, and projects categories stand out. In this sense, the use of multiple execution modes is the most studied job- and resource-based feature. Moreover, in studying resource-based features, non-renewable resources and the costs associated with resource use

[4] http://sbuweb.tcu.edu/tbrowning/RCMPSPinstances.htm
[5] http://www.eii.uva.es/elena/RCMPSPLIB.htm
[6] https://gent.cs.kuleuven.be/mista2013challenge/index.html

Table 13
Research works applied to real environments. Note (*) represents the average value.

Research	Instance				Objective functions								Features																			
					Rsrc.		Proj.	Time					Jobs				Jobs-rsrc.		Rsrc.								Proj.-rsrc.	Proj.		Relp.	
	Domain	# proj.	# jobs	# rsrc.	NPV	TPC	TPP	PQ	APD	WPD	M	WM	PP	JQ	R/L	JU	JCP	MM	SK	D	L	NR	CL	PRU	RR	RC	M/U	PCP	W	PS	C/N	GL	TL
Cui et al. (2021)	Manufacturing	4	288*	-							•								•														
Kannimuthu et al. (2020)	Construction	3	32	22		•		•			•			•				•								•				•			
Tian et al. (2020)	Aviation industry	3	6	1						•	•										•									•			
Song et al. (2018)	Production	5	7	6							•					•																	
Tian et al. (2018)	Undefined	12	30	27							•					•																	
Xin et al. (2018)	Heath Care	3	8	3								•															•				•		
Zhang & Chen (2018)	Construction	6	20	5		•					•							•			•	•				•				•			
Joo & Chua (2017)	Undefined	3	≥ 135*	≥ 20					•					•	•			•															•
Gholizadeh-Tayyar et al. (2016)	Construction	2	14	3		•																•	•			•						•	•
Tayyar et al. (2016)	Construction	2	14	3		•																•	•			•						•	•
Sonmez & Uysal (2015)	Undefined	3	46	11							•																						
Xu & Feng (2014)	Construction	3	18	2		•		•				•					•	•				•							•				
Xu & Zhang (2012)	Construction	6	6	3						•	•					•													•				
Chen & Shahandashti (2009)	Undefined	3	46	11							•																					•	
Gutjahr et al. (2008)	E-commerce	18	10	-			•												•									•			•		
Kolisch & Meyer (2006)	Pharmaceutical	100	-	11			•										•	•												•			
Chen (1994)	Maintenance	4	19	9		•					•											•				•		•			•		
Vercellis (1994)	Construction	10	20	6	•													•								•		•		•	•		
Speranza & Vercellis (1993)	Construction	10	20	6	•						•							•				•				•		•			•		

are prominent features. Lastly, the feature involving weights is the most used project-based feature.

Based on the reported works, we can observe that only a part of the objective functions found in the literature is pursued, where the makespan minimization and the total project cost minimization are the most used ones. It is worth indicating that the objective function related to project quality maximization was only used in real environments. The remaining objective functions do not exceed the 35% utilization in real problems. On the other hand, the consideration of weights, multiple execution modes, resource cost, and non-renewable resources are features addressed in many works, although they do not exceed 30% of the corresponding investigations carried out compared to the overall available literature. In addition, job quality, capacity limitations, planned unavailability, movable and unmovable resources, and generalized precedence relations are studied in few works but always considered in real environments.

8. Conclusions

This paper presented a review of RCMPSP evolution by analyzing related variants in terms of problem features, objective functions, proposed solution methods, benchmarks, and connection to practice. Also, taxonomies were proposed to unify concepts and visualize the current state of RCMPSP-related variants.

The conducted analysis showed that the most used assumption is to solve the problem under deterministic conditions. Despite this, the last decade presented a notable growth in the study of uncertainty associated with the job duration. Latest research lines mainly focus on studying decentralized environments, local and non-renewable resources, multiple execution modes, and different weights for each project. This led to the increase of decentralized solution methods such as multi-agent systems and combinatorial auctions and the development and use of the MPSPLib library as a common benchmark to validate and compare results. On the other hand, time- and resource-based objective functions were the most studied, more specifically with the minimization of the makespan, average project delay, and total project cost being the most predominant objectives. In addition, the approximate algorithms were the most used for the resolution of the RCMPSP-related problems, highlighting the use of genetic algorithms and priority rule-based algorithms. It is also worth noting the significant increase in the use of hybrid algorithms for the resolution of the proposed variants.

The works applied to real environments were analyzed. Some indications are given concerning aspects that can be studied in greater depth to promote the use and application of RCMPSP as well as reduce the gap between theory and practice. For instance, even though many processes are automated, company staff represents most resources in project management, generating high complexity levels in schedule generation. Given this, increasing the study of different skills and performance of a worker compared to the different jobs that staff can execute can improve production processes. Furthermore, the consideration of uncertainty in production processes is an aspect slightly addressed in practice, only associated with job duration, so it is necessary to widen the scope of future studies in this respect. Concerning objectives, the most used ones are those aligned with economic aspects and/or project completion times without emphasizing the resource use that generates nature-friendly production processes. In this sense, it may be relevant to increase the study of non-renewable and partially renewable resources while pursuing objectives such as reducing resource consumption or even reducing and leveling any resource within the time horizon. Furthermore, studying and promoting the use of objectives such as minimizing idle resources, or inventory in processes in real environments, can allow efficient resource

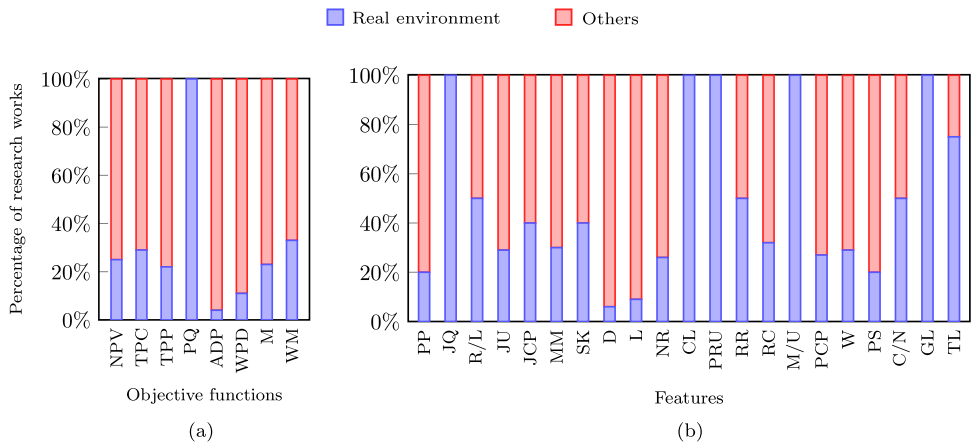

Fig. 14. Relationship between the literature and the real environments collected in this survey.

use. Also, increasing different precedence relationships where common resource use is made can considerably reduce resource use.

Regarding solution methods to satisfy current production processes, one of the main factors that must be provided is the resilience capacity to schedule changes, especially under uncertain conditions. Furthermore, despite recent work, the increased use of solution methods based on multi-agent systems and combinatorial auctions, as well as hybrid algorithms, recommend a valuable line of research for further study. In this sense, an interesting research direction is the in-depth study of how the specific multi-project features are handled in the algorithms used in the literature to solve the different variants of the RCMPSP. Finally, the evaluation and comparison of the performance of proposed algorithms are essential for RCMPSP progress, so the creation of public libraries valuing the related features studied in the literature is essential for promoting and fostering the study of new problem variants.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A.1
List of acronyms

Acronym	Definition
APD	Average project delay
APE	Average project earliness
APPD	Average percentage project delay
APT	Average performance time
Avg.	Average
B	Blocking

(*continued on next page*)

Table A.1 (*continued*)

Acronym	Definition
C/N	Concurrency or non-concurrency
CL	Capacity limitations
CP	Constraint programming
D	Decentralized
DP	Dynamic programming
DR	Detection and rework
EX	Exclusiveness
F	Flexible
GA	Genetic algorithms
GL	Generalized
IPI	In-process inventory
IR	Idle resources
JCP	Job cost/profit
JQ	Job quality
JS	Job selection
JU	Job uncertainty
L	Local
LP	Location pool
LP_b	Linear programming based algorithms
M	Makespan
M/U	Movable or un-movable
MAS-CA	Multi-agent systems and combinatorial auction
MM	Multiple modes
MNPG	Multi-network problem generator
MPSPLib	Multi-project scheduling problem library
NPV	Net present value
NR	Non-renewable
OA	Other algorithms
PCP	Project cost/profit
Pct.	Percentage
Perf.	Performance
PP	Preemption
PQ	Project quality
PR_b	Priority rules based heuristics
Proj.	Project
PRR	Partially renewable
PRU	Planned resource unavailability
PS	Project selection
PSP	Project splitting
PSPLib	Project scheduling problem library
PU	Project uncertainty
R	Robustness
R/L	Release/latest date
RC	Resource cost
RC	Resource cost
RCMPSP	Resource-constrained multi-project scheduling problem
RCMPSPLIB	Resource-constrained multi-project scheduling problem library
RCPSP	Resource-constrained project scheduling problem
Relp	Relationship
RL	Resource leveling

(*continued on next page*)

Table A.1 (*continued*)

Acronym	Definition
RR	Resource rent
RRNTS	Required resource for a non-tardy schedule
Rsrc.	Resources
RU	Resource uncertainty
SA	Simulated annealing
SD	Standard deviation
SDML	Standard deviation mean lateness
SI	Swarm intelligence
SK	Skills
TB	Time balance
TC	Transfer cost
TL	Time lags
TPC	Total project cost
TPP	Total project profit
TS	Tabu Search
TT	Transfer time
VR	Variable request
W	Weight
WM	Weighted makespan
WPD	Weighted project delay
WTD	Weighted total delay
Z	Zone

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